Markscheme

Mathematics: analysis and approaches
Practice question bank

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97. (a)

It can be proven by induction that $u_n = 3(2^n) - 1$ satisfies the recurrence $u_1 = 5$, $u_{n+1} = 2u_n + 1$, for every positive integer n.

Using this result, find the value of u_6.

substitute $n = 6$: $u_6 = 3(2^6) - 1$ *(M1)*

$= 3(64) - 1$ **A1**

$= 191$ **A1**

[4 marks]